

# Introduction to Inference

CSU, Chico Math 314

2025-10-06

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## Review: sampling distribution

$\bar{X}$  consists of and therefore is itself a random variable. The sampling distribution for  $\bar{X}$  is often unknown, but reasonably well approximated by the normal distribution due to the Central Limit Theorem<sup>1</sup>.

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<sup>1</sup>Assumptions, some quite plausible, must be met for the CLT to apply, but in practice these assumptions are often met.

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# Towards Interval Estimation

The hope is that an interval, rather than a point estimate, will better capture the true parameter of interest.

If we want to be *certain*, why don't we just take a 100% Confidence Interval?

(HINT: Just because a confidence interval contains the true mean doesn't mean it's actually useful!)

# Confidence Intervals, idea

We will use the sampling distribution of  $\bar{X}$  to help us make informed decisions about the population mean. Instead of throwing a dart, we will cast a net, with bounds determined by the **margin of error** (ME), discussed in the next few slides.

$$[\bar{X} - ME, \bar{X} + ME].$$

## Confidence Interval, example (to convey idea)

A sample from the Youth Risk Behavioral Surveillance System (YRBSS) suggests the average student height is  $\bar{X} = 1.697$  meters. Construct a confidence interval.

(Note: the "0.15" is arbitrary here—we'll look at how to choose the margin of error later in the notes.)

$$[1.697 - 0.15, 1.697 + 0.15]$$



# Confidence Interval, definition

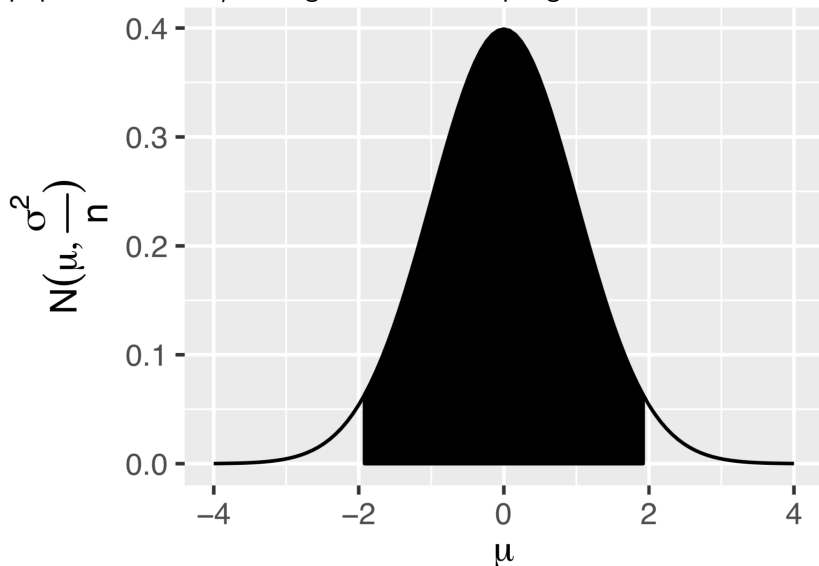
Our net is called a **confidence interval**, and we make a compromise between width and confidence.

## confidence interval

A confidence interval is an interval  $[\hat{\theta}_L, \hat{\theta}_U]$  computed from a sample, used to estimate a population parameter of interest.

## Confidence Interval, a trade off

We will calculate  $100(1 - \alpha)\%$  confidence intervals about the population mean  $\mu$ , using  $\bar{X}$  and its sampling distribution.



## Confidence Intervals, margin of error

To create an interval we add and subtract some amount from  $\bar{X}$ . This amount is the **margin of error**.

### margin of error

In a confidence interval, the margin of error is  $z_{1-\alpha/2}^* \cdot \sigma_{\bar{X}}$ , where  $\sigma_{\bar{X}}$  is the standard error and  $z_{1-\alpha/2}^*$  is a constant that is determined by the value of  $\alpha$  as follows:

$$z_{1-\alpha/2}^* = qnorm(1 - \frac{\alpha}{2})$$

**NOTE:** This constant is sometimes referred to as a **critical value**. Note again that it is a **constant** fixed according to  $\alpha$ , and is separate from the  $z$  seen for hypothesis testing in the next section of the slides.

## Confidence Interval, sample mean

Assume  $\bar{X} \sim N(\mu, \sigma_{\bar{X}}^2)$  such that  $\sigma_{\bar{X}}^2 = \text{Var}(\bar{X}) = \frac{\sigma^2}{n}$ . A confidence interval<sup>2</sup> is constructed as

$$[\bar{X} - z_{1-\alpha/2}^* \cdot \sigma_{\bar{X}}, \bar{X} + z_{1-\alpha/2}^* \cdot \sigma_{\bar{X}}].$$

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<sup>2</sup>If  $\alpha = 0.05$ , then a 95% confidence interval will use  $z_{1-\alpha/2}^* = \text{qnorm}(0.975) \approx 1.959964$ . See slide 10 for visual representation.

## Confidence Intervals, literal translation

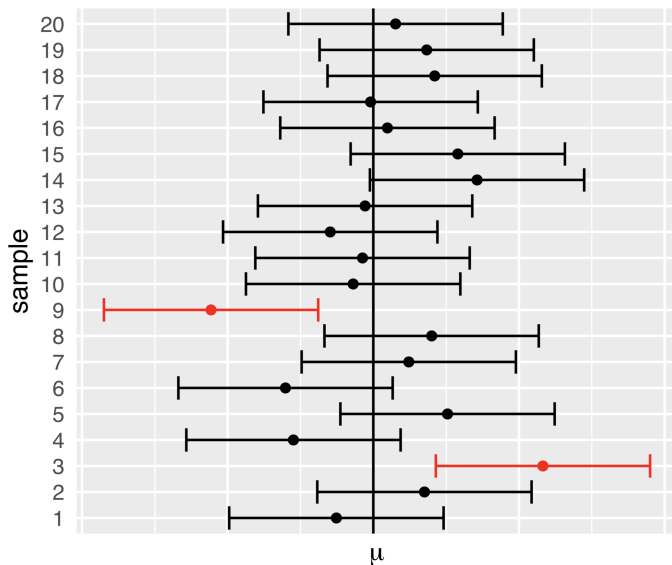
If we were to re-sample  $N$  times and create a confidence interval from each new sample (of size  $n$ ),  $100(1 - \alpha)\%$  of those intervals would include the true population mean<sup>3</sup>.

Beware, a 95% confidence interval does NOT have a 95% probability of containing the true mean! It either does or it doesn't (see footnote)!

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<sup>3</sup>Confidence intervals say nothing about individual observations. They only make statements about the population parameters we are estimating.

# Confidence Intervals, translation via picture



## Confidence Interval, example

A sample,  $n = 100$ , from the Youth Risk Behavioral Surveillance System (YRBSS) suggests the average student height is  $\bar{X} = 1.697$  meters with a standard deviation of  $\sigma = 0.088$  meters. What is an approximate 99% confidence interval for the average height of all of the YRBSS students?

$$[1.697 - 2.58 \cdot 0.088/\sqrt{100}, 1.697 + 2.58 \cdot 0.088/\sqrt{100}]$$

Note that here,  $z_{1-\alpha/2}^* = \text{qnorm}(0.995) \approx 2.58$ .

## Confidence Interval, example in R

A sample,  $n = 100$ , from the Youth Risk Behavioral Surveillance System (YRBSS) suggests the average student height is  $\bar{X} = 1.697$  meters with a standard deviation of  $\sigma = 0.088$  meters. What is an approximate 99% confidence interval for the average height of all of the YRBSS students?

```
xbar <- 1.697           # sample mean
n <- 100                # sample size
alpha <- 0.01           # 99% = 1 - 0.01
se <- 0.088/sqrt(n)     # standard error
z.star <- qnorm(1-alpha/2) # critical value
ME <- se * z.star       # margin of error
c(xbar - ME , xbar + ME) # confidence interval (CI)

## [1] 1.674333 1.719667
```



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# Motivating Hypothesis Testing Framework

Suppose your buddy claims he shoots 90% from the free throw line. Since we are all statisticians, we a) don't believe him, and b) insist upon testing his claim empirically. So we collect some data. He steps up and starts shooting. At what point do we reject his claim?

# Motivating Hypothesis Testing Framework

Implicitly, we used logical framework to evaluate our buddy's claim. Let's unpack that framework and give it a name.

- ▶ Established two hypotheses
  - ▶ he shoots as well as he says he does,  $p = 0.9$
  - ▶ he does not shoot as well as he claims,  $p < 0.9$ .
- ▶ Collected data
  - ▶ He took  $n$  shots from the line.
- ▶ Analyzed the data
  - ▶ Estimated  $p$  with  $\hat{p}$
  - ▶ Determined the probability of observing  $\hat{p}$  if indeed he shoots as well as he claims.
- ▶ Made a conclusion based on your analysis
  - ▶ Given his claim, if  $\hat{p}$  seems too unlikely, it's probably not true.

# Hypothesis Test, example

A trucking firm suspects that a spark plug manufacturer is lying about the average lifetime of its spark plugs. The manufacturer, Sparky, claims their spark plugs last 28,000 miles on average. From a sample of 27 spark plugs, we find  $\bar{X} = 26,040$  with standard deviation  $\sigma = 12,894$ . Test Sparky's claim about the average lifetime of their spark plugs.

- ▶ Establish hypotheses.
- ▶ Collect data.
- ▶ Analyze data.
- ▶ Make conclusion.

## Hypothesis Test, example

A trucking firm suspects that a spark plug manufacturer is lying about the average lifetime of its spark plugs. The manufacturer, Sparky, claims their spark plugs last 28,000 miles on average. From a sample of 27 spark plugs, we find  $\bar{X} = 26,040$  with a standard deviation of 12,894. Test Sparky's claim about the average lifetime of their spark plugs.

- ▶  $\mu = 28,000$  versus  $\mu < 28,000$
- ▶  $z = \frac{\bar{X} - \mu}{\sigma_{\bar{X}}} = \frac{26,040 - 28,000}{12,894 / \sqrt{27}}$
- ▶  $P(Z < z) = 0.21$
- ▶ Given data, safer bet seems to decide  $\mu = 28,000$ .

**NOTE:** The  $z$  seen here is often called a **z-score**, and is separate from the  $z^*$  seen with confidence intervals. This  $z$  is obtained by **standardizing** the sample mean.

# Hypothesis Testing Framework

We call this framework **hypothesis testing**. Let's rephrase hypothesis testing into the language of statistics.

- ▶ Establish hypothesis, null ( $H_0$ ) and alternative ( $H_1$ ).
  - ▶  $H_0$  and  $H_1$  are statements about population parameters
- ▶ Collect data.
- ▶ Analyze data (calculate summary statistics) – we'll most often be estimating  $\mu$  – and **p-value**.
- ▶ Make conclusion by comparing p-value to **level of significance**.

## p-value, definition

The **p-value** helps us decide between  $H_0$  and  $H_1$ .

### p-value

The probability of observing the test statistic we did, or something more extreme<sup>4</sup>, assuming the null hypothesis is true.

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<sup>4</sup>smaller, larger, or both, depending on the alternative hypothesis.

## level of significance

We define a largest level at which we are willing to incorrectly conclude. We call this value the level of significance, and give it the symbol  $\alpha$ <sup>5</sup>.

### level of significance

The largest probability of incorrectly rejecting  $H_0$  when in fact  $H_0$  is true.

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<sup>5</sup>Common values of  $\alpha$  are 0.05 and 0.01, but these are not the only possibilities.



# hypotheses

The null and alternative hypotheses generally follow some conventions.

- ▶  $H_0$  declares the parameter of interest to be equal to some value.
  - ▶  $H_0 : \mu = 10$
- ▶  $H_1$  declares the (same) parameter of interest to be less than, greater than, or not equal to the same value in the null hypothesis  $H_0$  – the researcher chooses one before conducting the test.
  - ▶  $H_1 : \mu \{<, >, \neq\} 10.$

## Hypothesis Test, another example

Suppose a tech company wants to test a novel web browser. They sample 1014 randomly selected websites and make a simple decision: this website was displayed properly on the browser or it was not. They found that 984 websites displayed correctly. Test whether or not the browser displays 99% of web pages correctly, and compare your conclusion to a confidence interval. Choose  $\alpha = 0.05$ .

- ▶ Establish hypotheses:  $H_0$  and  $H_1$ .
- ▶ Collect data.
- ▶ Analyze data  $\Rightarrow$  calculate summary statistic and p-value.
- ▶ Make conclusion  $\Rightarrow$  decide  $H_0$  or  $H_1$  via p-value ?  $\alpha$

# Hypothesis Test, another example

Hypothesis Test:

$$H_0 : p = .99 \quad \text{versus} \quad H_1 : p \neq .99$$

NOTE: When using a hypothesis test, the value of  $p$  in the null hypothesis should be used to calculate  $\sigma$ .

```
n <- 1014
p <- 0.99 # Population parameter according to H_0
phat <- 984/n # Estimate for the population parameter
sigma <- sqrt(p*(1-p)) # St. dev for Bernoulli variable
z <- (phat - p)/(sigma/sqrt(n)) # assume H_0 true
2*(1 - pnorm(abs(z))) # compare to alpha = 0.05

## [1] 1.826257e-10
```

## Hypothesis Test, another example

100(1 - 0.05)% Confidence interval.

NOTE: A confidence interval doesn't have a value of  $p$  to reference when calculating sigma. Use  $\hat{p}$  instead.

```
n <- 1014
phat <- 984/n
alpha <- 0.05
sigma <- sqrt(phat*(1-phat))
se <- sigma/sqrt(n)
z.star <- qnorm(1-alpha/2)
ME <- se * z.star
c(phat - ME , phat + ME)

## [1] 0.9599850 0.9808434
```

# Comparing the hypothesis test and confidence interval

Since the p-value is less than our level of significance  $\alpha = 0.05$ , we **reject**  $H_0$ , concluding that a proportion of pages other than 0.99 are displayed correctly.

The equivalent (95%) 2-sided confidence interval did **not** include the null hypothesis value of  $p = 0.99$ . This is in line with our rejection of the null hypothesis in our hypothesis test.

## What about one-sided hypothesis tests?

The hypothesis test above may have been better designed as a one-sided hypothesis test, with  $H_1 : p < .99$ .

However, a two-sided confidence interval can't be used to describe a one-sided hypothesis test—we will instead need to consider a **one-sided confidence interval** to make a direct comparison.

We'll see examples of how to set these up in the upcoming Examples PDF.

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Adapted from notes by Dr. Roualdes.